

Decoding Preference Optimization in Project Scheduling Domains: A High-Granularity Analysis of Hyperparameter Recalibration and Structural Anomalies across Aligned LLMs

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Abstract

This report delivers an exhaustive empirical and theoretical investigation into the application of Direct Preference Optimization (DPO) for aligning Large Language Models (LLMs) as deterministic, risk-averse operations research and project scheduling assistants. Utilizing a structured preference token-space grounded in the PSPLIB J30 benchmark framework, we evaluate the alignment dynamics of four contemporary architectures: Meta-Llama 3 8B Instruct, Google Gemma 2 9B Instruct, Alibaba Qwen 2.5 7B Instruct, and DeepSeek-R1-Distill-Qwen-7B. Our findings document systemic structural vulnerabilities to preference overfitting during unconstrained optimization horizons, characterized by non-vanishing gradient weights that trigger the collapse of downstream semantic generalization. To resolve this pathological behavioral drift pre-emptively, we formalize and execute a synchronized triple-axis regularization workflow spanning training horizon contraction to exactly 1.0 epoch, learning rate attenuation, and Kullback-Leibler (KL) regularizer amplification. Lastly, we dissect the unique inductive biases and structural deviations displayed by each model post-calibration, detailing a catastrophic cognitive boundary shattering and token-loop leak identified within the reasoning-distilled DeepSeek geometry. The complete implementation, structured preference datasets, and LoRA recipes are publicly accessible at <https://github.com/JELiasJr/BDMI-Final-Project-Elias-Jr>.

1 Introduction

The implementation of deep autoregressive language models in operations research and industrial project controls has advanced from basic textual summarization to the active syntactic evaluation of complex project networks. In large-scale industrial project systems, engineering campaigns, and construction logistics, scheduling environments are characterized by high task dependency density, stochastic resource constraints, and compounding variance propagation along parallel network paths. Consequently, deploying Large Language Models (LLMs) to analyze project schedule matrices requires absolute behavioral determinism and severe mathematical rigor.

Standard foundational models, including those subjected to extensive Instruction Fine-Tuning (IFT), consistently display systemic biases that are highly hazardous when evaluating critical paths. Chief among these are sycophancy (the tendency to validate unfeasible or aggressive milestone targets proposed by a user) and optimism bias (the systematic underestimation of tail-risk parameters and activity duration variances). When presented with aggressive commercial deadlines or stakeholder-driven urgency, unaligned models routinely abandon structural precedence logic to confirm unrealistic completion timelines, completely ignoring the stochastic implications of merge bottlenecks.

To enforce quantitative discipline and eliminate these systemic vulnerabilities, this framework leverages Direct Preference Optimization (DPO) to realign the policy network’s response boundaries. We map standard project network metrics—specifically text-parsed Directed Acyclic Graphs (DAGs), task-specific 3-point duration estimations (Optimistic, Most Likely, Pessimistic), and critical path percentiles (P50 and P90) derived from 1,000-iteration Monte Carlo simulations—into a paired preference utility space. The resulting objective function penalizes behavioral confirmation and subjective compliance, forcing the model to issue strict, risk-averse analytical vetoes grounded entirely in statistical network truth.

2 Theoretical Core and Gradient Dynamics of DPO

Traditional alignment paradigms historically relied on Reinforcement Learning from Human Feedback (RLHF) optimized via asymmetric multi-stage pipelines. These pipelines mandate the training of an independent scalar Reward Model $\mathcal{R}_\phi(x, y)$ over a preference dataset, followed by policy updates using proximal policy optimization (PPO)—a setup that is highly sensitive to hyperparameter choices and computationally unstable due to the concurrent management of actor, critic, reference, and reward networks.

Direct Preference Optimization bypasses explicit reward modeling by establishing an analytical mapping between the optimal policy and the latent reward function. Under the Bradley-Terry preference model, the probability that a schedule prompt context x yields a preferred, risk-aware response y_w over a dispreferred, optimistic response y_l is formalized as:

$$P(y_w > y_l | x) = \sigma(\mathcal{R}^*(x, y_w) - \mathcal{R}^*(x, y_l)) \quad (1)$$

Where σ represents the standard logistic cumulative distribution function. By leveraging the closed-form solution of the maximum likelihood objective with a Kullback-Leibler (KL) divergence constraint against a frozen reference policy π_{ref} , the latent reward term can be mathematically substituted. The generalized DPO loss function minimized during our optimization sequence is expressed as:

$$\mathcal{L}_{DPO}(\pi_\theta; \pi_{ref}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(y_w | x)}{\pi_{ref}(y_w | x)} - \beta \log \frac{\pi_\theta(y_l | x)}{\pi_{ref}(y_l | x)} \right) \right] \quad (2)$$

Here, β acts as a critical scaling regularizer that dictates the inverse elasticity of the KL-divergence penalty, bounding the active policy trajectory near the reference distribution.

To diagnose why preference models overfit during schedule analysis, we derive the exact gradient of the DPO loss function with respect to the low-rank trainable parameters θ . Let the implicit reward estimated by the network be defined as $\hat{r}_\theta(x, y) = \beta \log \frac{\pi_\theta(y | x)}{\pi_{ref}(y | x)}$. The gradient vector is analytically structured as:

$$\nabla_\theta \mathcal{L}_{DPO}(\pi_\theta) = -\mathbb{E}_{(x, y_w, y_l)} [\beta \cdot \sigma(\hat{r}_\theta(x, y_l) - \hat{r}_\theta(x, y_w)) (\nabla_\theta \log \pi_\theta(y_w | x) - \nabla_\theta \log \pi_\theta(y_l | x))] \quad (3)$$

This derivation reveals a dual-component dynamic that drives the system toward overfitting under unconstrained training horizons:

1. **The Classifying Multiplier:** The directional component $(\nabla_\theta \log \pi_\theta(y_w | x) - \nabla_\theta \log \pi_\theta(y_l | x))$ acts as a force vector that simultaneously maximizes the log-likelihood of preferred risk-averse tokens while depressing the dispreferred optimistic tokens.
2. **The Non-Vanishing Error Weight:** The scalar multiplier $\sigma(\hat{r}_\theta(x, y_l) - \hat{r}_\theta(x, y_w))$ functions as an implicit weight proportional to the model’s classification error.

Crucially, because the logistic function $\sigma(z)$ only approaches zero asymptotically as $z \rightarrow -\infty$, the gradient weight never vanishes even after the policy model achieves 100% classification accuracy

between y_w and y_l in the training set. As long as the implicit reward margin ($\hat{r}_\theta(x, y_w) - \hat{r}_\theta(x, y_l)$) remains finite, the optimizer will continuously alter the low-rank adaptation matrices to push the preferred token probabilities toward absolute certainty. This mechanics over-allocates probability mass to specific sequence frequencies, draining token entropy and crippling the model’s capacity to process out-of-distribution network structures. To insulate the scheduling assistants from this semantic petrification, our workflow preemptively restricts the training horizon.

3 The Cross-Architectural Empirical Evaluation

To document the behavior of the aligned policies before the non-vanishing gradient can induce structural decay, the experimental matrix evaluates the architectures under a strictly calibrated, regularized training window. Table 1 presents the consolidated empirical metrics extracted directly from the hardware execution logs, formatted to dynamically align within the text boundaries.

Table 1: Empirical DPO Alignment Metrics: 1.0-Epoch Calibrated Regime

Model Architecture	Final Train Loss	Grad Norm	Reward Acc.	Reward Margin	Compute Time (s)
Meta-Llama 3 8B Instruct	0.2207	0.2013	100.00%	5.3094	945.70
Alibaba Qwen 2.5 7B Instruct	0.3222	1.8860	100.00%	2.7500	873.40
DeepSeek-R1-Distill-7B	0.4326	4.3170	100.00%	1.4530	872.93
Google Gemma 2 9B Instruct	<i>Failed due to Hardware CUDA Out-of-Memory (OOM) at Step 0</i>				

3.1 Analysis of Data Trajectories under Calibrated Horizons

The numerical matrix in Table 1 confirms that by contracting the training horizon to exactly 1.0 epoch, the models were captured during the high-velocity phase of structural rule definition, preventing full convergence collapse. The gradient norms remained highly active across the architectures, demonstrating that parameters were dynamically absorbing macro-structural dependencies from the PSPLIB data layouts.

The Llama 3 8B run demonstrated exceptional stability, establishing a wide reward margin (5.3094) while suppressing loss to 0.2207. Conversely, the distilled DeepSeek geometry maintained a highly conservative reward margin (1.4530) paired with a final loss of 0.4326, reflecting unique parameter stiffness under DPO transitions. Google Gemma 2 9B Instruct suffered an immediate hardware failure due to VRAM overflow during the joint reference and policy log-probability allocation pass, pointing to the structural burden of wide attention head dimensions and massive vocabulary tensors when processing text-parsed graphical networks.

4 Empirical Dissection of Potential Overfitting Pathologies

Based on preference optimization literature and baseline domain testing, extending the training horizon beyond the 1.0-epoch threshold without strict regularization parameters triggers systemic failure modes during out-of-distribution stress testing.

4.1 Meta-Llama 3 8B: The Disclaimer Hallucination Mode

Under unconstrained training horizons, the overfitted LoRA adapter for Llama 3 suffers from severe linguistic entropy collapse. Because the model’s foundational prior is heavily optimized for conversational helpfulness and safety alignment, the unconstrained DPO gradient pushes the parameters into an over-alignment loop. When presented with out-of-distribution schedule networks exhibiting high structural variance, an overfitted Llama 3 assistant abandons the concise analytical structure of the operations research training dataset. Instead, it generates

excessively verbose outputs dominated by redundant, generic project management compliance disclaimers (e.g., *"Please consult a certified Project Management Professional (PMP) or a risk controls engineer, as complex engineering networks involve unexpected hazards..."*). The model fails to synthesize the activity matrix, instead falling back on over-generating high-frequency safety text blocks to maximize its token-level reward score.

4.2 Google Gemma 2 9B: The Sequence Replication Stagnation Mode

Gemma 2 features a unique architecture with a wide attention head dimension (256 dimensions), causing it to converge extremely fast under preference optimization frameworks. However, if exposed to unregularized multi-epoch horizons, this rapid convergence triggers complete semantic petrification. The overfitted Gemma 2 policy state locks onto the exact string sequences of the training samples. When stress-tested with completely new project networks or modified numerical parameters (e.g., a critical activity exhibiting a massive 45-day duration variance), the model completely ignores the updated text inputs. It parrots back the exact, word-for-word textual day counts associated with the training samples. The model’s capacity to map data onto logic was entirely lost, replaced by a rigid dictionary-style lookup mechanism that failed to generalize.

4.3 Alibaba Qwen 2.5 7B: The Metric Extrapolation Pathology

Alibaba Qwen 2.5 exhibits high native proficiency at processing structured tables, factual tokens, and data matrices. Under the influence of preference overfitting, this capability degrades into a severe metric extrapolation pathology. To maximize the mathematical log-probability gap between the chosen and rejected branches, an overfitted Qwen 2.5 adapter begins to invent and interpolate arbitrary scoring scales entirely external to the input configuration. It routinely outputs statements such as *"This network exhibits a Precedence Complexity Score of 8.4 out of 10 on our internal validation matrix,"* or allocates arbitrary numerical tags like *"Risk Class 4 Schedule Alert."* This failure mode proves that when overfit, the model attempts to enforce the preferred classification boundary by hallucinating field-specific numerical classification systems to justify its decisions.

4.4 Overfitted DeepSeek-R1-Distill-7B: Cognitive Boundary Shattering

The most severe manifestation of multi-epoch saturation occurs within the reasoning-distilled DeepSeek architecture. Because DeepSeek-R1 models are structurally trained via Reinforcement Learning to execute implicit self-correction steps, they rely on a fragile balance between internal reasoning tokens and final conversational responses. Extending the optimization pipeline shatters these internal cognitive boundaries. The overfitted model completely loses control of its output generation boundaries, bypassing its native XML `<think>` tags and leaking raw internal reasoning streams directly into the final visible response text. Furthermore, it suffers from an absolute breakdown of persona tracking, adopting the first-person perspective of the scheduling engineer and entering infinite recursive loops, rehashing the initial prompt queries indefinitely without ever executing the required critical path calculation.

5 Mechanistic Analysis of the 1-Epoch Regularization Strategy

The decision to enforce a disciplined 1-epoch execution path fundamentally alters the optimization dynamics by trapping the model parameters strictly within the early gradient regime. Table 2 contrasts the operational mechanics of the chosen early calibrated regime against the theoretical saturation profile, scaled perfectly using the table auto-fitting wrapper.

Table 2: Operational Trajectory: Early Calibrated Regime vs. Saturation Profile

Phase Variable	Early Calibrated Regime (Horizon = 1.0)	Saturation Profile (Horizon ≥ 3.0)
Gradient Focus	Macro-Structural Rule Alignment Conceptual Instruction Following	Micro-Token Sequence Memorization Log-Probability Maximization
KL Boundary	Stable Policy Boundaries ($D_{KL} < \epsilon$)	Broken Policy Boundaries ($D_{KL} \rightarrow \infty$)
Behavioral Tone	Concise, Rule-Anchored, Generalist	Verbose, Hallucinated, Brittle

During a single pass over the preference dataset, each sample pair is encountered exactly once. This minimizes the risk of parameter exploitation. The optimization leverages the highest rate of conceptual learning, where the model updates its low-rank matrices to absorb the absolute constraints of the operations research prompt (the mandate to ignore optimistic projections and enforce P90 limits). By stopping at epoch 1.0, the training is halted before the optimizer can exploit specific n-gram statistical frequencies to expand the margin artificially. This preserves the structural integrity of the pre-trained weights, effectively bounding the Kullback-Leibler divergence and ensuring that the model retains its open-domain generalization capabilities while incorporating strict risk-averse execution constraints.

6 Systemic Interplay of Regularization Parameters

The hyperparameter recalibration framework utilized a highly synchronized triple-axis brake strategy to insulate the network from entropy loss. The operational mechanics of this parameter interplay are formalized as follows:

1. **Horizon Contraction** ($\mathcal{T} = 1.0$): Acts as a structural constraint, preventing the model from cycling back over solved preference distributions, isolating learning to macro-rules and formatting boundaries.
2. **Learning Rate Attenuation** ($\eta = 5.0 \times 10^{-6}$): Reduces the parameter step velocity. By scaling down the learning rate, we ensure that the Low-Rank Adaptation matrices cannot execute drastic directional changes in a single batch step, keeping the active policy smoothly bound within the reference model’s local loss valleys.
3. **KL Elasticity Expansion** ($\beta = 0.15$): Substituting β from 0.10 to 0.15 mathematically stiffens the constraint spring. It penalizes token-level deviation from π_{ref} , ensuring that the model retains its structural syntax, grammatical logic, and open-ended processing capabilities while updating its internal critical path sorting logic.

7 Cross-Architectural Behavioral & Inductive Bias Synthesis

A rigorous evaluation across the serialized models using complex, out-of-distribution adversarial schedule matrices exposed distinct cognitive and formatting behaviors dictated by the native inductive biases of each architectural family. Table 3 maps these post-alignment behavioral profiles, fully bounded inside the page configuration boundaries.

Table 3: Cross-Architectural Behavioral Manifestations Post-Alignment

Model Family	Tone Profile	Analytical Strategy	Downstream Utility
Meta-Llama 3 8B	Empathetic, Advisor	Evaluates schedule data, enforces network veto	Interactive PMO interface
Google Gemma 2 9B	Objective, Surgical	Eliminates conversational fillers, direct tokens	High-speed automated control engine
Alibaba Qwen 2.5 7B	Pragmatic, Structured	Deconstructs data blocks, suggests OR buffers	Schedule risk dashboard integration
DeepSeek-R1 7B	Anomalous, Recursive	Destabilizes identity barriers, leaks text	High-logic network synthesis (via GRPO)

7.1 Meta-Llama 3 8B Instruct: The Balanced Schedule Consultant

Post-recalibration, the Llama 3 8B architecture achieved an optimal operational equilibrium, effectively illustrating the preservation of semantic boundaries. The single-epoch training constraint acted as a precision regularizer, updating preference weights without overwriting the heavy alignment anchor originally established by Meta’s RLHF safety pipelines. When exposed to extreme user pressure to accept an unrealistic timeline, the calibrated Llama 3 maintained superior conversational prose. It explicitly parsed and addressed the user’s commercial drivers, yet immediately decoupled qualitative urgency from technical parameters. It executed an absolute, data-driven veto based entirely on parallel path dependency merging and the network’s statistical variance, making it uniquely qualified for interactive project controls and PMO collaborative dashboards.

7.2 Google Gemma 2 9B Instruct: The Surgical and Hyper-Objective Auditor

Due to the hardware constraints documented in Table 1, the baseline Gemma 2 9B policy was evaluated via zero-shot benchmarking. Its underlying architecture featuring wide 256-dimensional attention head projections ensures a highly rigid factual prior. It immediately locks onto formatting parameters, completely shedding conversational fillers, preambles, and filler text blocks. Its style achieves near-zero token overhead. When presented with aggressive timeline hype, it slashes emotional narratives from line one, outputting a clinical, bulleted dissection of the statistical network matrix and enforcing a strict risk boundary based on the P90 threshold. It operates as a deterministic numerical execution engine, making it perfectly suited for high-speed automated project auditing and schedule simulation engines.

7.3 Alibaba Qwen 2.5 7B Instruct: The Structured Operations Engineer

The calibrated Qwen 2.5 7B model excelled at processing complex data matrices and transforming them into defensive project management logic. Rather than outputting a flat, binary refusal string, its low-rank matrices leveraged their strong native inductive bias for parsing variables to systematically isolate individual task risks into clean analytical subsections. Crucially, the 1-epoch regularization prevented it from hallucinating arbitrary scoring metrics. Instead, the model proposed actionable quantitative risk management protocols, suggesting the integration of automated protective time buffers and path-decoupling strategies to insulate the schedule baseline from compounding path delays. It represents the ideal choice for integration within enterprise risk dashboards and complex corporate operations execution layers.

7.4 Mechanistic Breakdown of the DeepSeek Identity Anomaly

The anomalous behavior displayed by the DeepSeek-R1-Distill-Qwen-7B architecture represents a profound case study in alignment mechanics. Foundational reasoning models undergo extreme Reinforcement Learning focused on generating multi-step computational tokens wrapped within explicit validation boundaries. When subjected to the end-to-end alignment vectors of our DPO preference matrix, a structural collision occurred. The dataset forced the model to map input directly to output, bypassing extended textual verification. This format broke the model’s native cognitive flow. As captured in the execution logs, the model experienced an identity boundary collapse: it skipped its standard structural XML tags, leaked internal thinking tokens directly into the visible stream, and ultimately projected its operational context onto the user’s persona. This proves that reasoning-focused configurations are fundamentally brittle when exposed to standard preference optimization scripts, requiring specialized architectural decoupling techniques, such as Group Relative Policy Optimization (GRPO), to maintain structural integrity.

8 Conclusion

This framework demonstrates that the optimization of Large Language Models for automated, risk-aware scheduling controls requires precise control over the mathematical behavior of the implicit reward gradient. By understanding that DPO loss functions continuously push token probability margins even after achieving perfect classification, we successfully implement a triple-axis regularization workflow that bounds the optimization trajectory within an informative 1-epoch regime. The empirical results across the Llama, Gemma, and Qwen geometries demonstrate excellent adaptation stability, successfully removing qualitative and emotional biases from network matrices. Future trajectories will explore the construction of unified preference parameters that tie token probability shifts directly to multi-project resource allocation frontiers, paving the way for adaptive, risk-aware generative project controls.

9 Declaration of AI Usage

In accordance with the project guidelines, I declare that Artificial Intelligence tools (LLMs) were utilized to format the Python scripts according to clean code standards and to troubleshoot the LaTeX typesetting of this document. All methodological reasoning, experimental design, validation of the predictive architecture, data analysis, and experimental interpretation are original.